

API Integration Manual: Premium SMS Service

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1. Introduction

This manual provides comprehensive details on integrating with Pure Bros HTTP interfaces for the management of Premium SMS Services. It includes technical specifications and procedural guidelines required to ensure a seamless API implementation.

Document Revision History

Version	Key Changes	Date
1.0	Initial version	-
2.0	Removal of obsolete interfaces	15/03/2008
2.2	New layout implemented	28/03/2009
3.0	Integration with PosteMobile	08/11/2010
3.3	General review	10/10/2011
3.4	General review	19/06/2012
3.5	General review	18/11/2015
3.6	General review	12/12/2024

2. Overview

The Italian Value-Added Services (VAS) market operates with two primary service types:

1. **Trigger Services (PUSH)** - Subscription-based services where the customer is enrolled in a database and billed at a specified frequency (daily, weekly, monthly, etc.).
2. **On-Demand Services (PULL)** - Services purchased by the customer on a one-time basis, without requiring a subscription.

To launch an SMS Premium service in the Italian market, a Service Description Document must be produced. This document outlines all aspects of the service, including activation and deactivation syntax, pricing, delivery frequency, and other regulatory requirements. It must be approved by Pure Bros and, in some cases, the mobile network operator (MNO).

3. Glossary

Acronym	Description
CC	<i>Country Code</i>
MSISDN	<i>Mobile Subscriber Integrated Services Digital Network Number</i>
NDC	<i>National Destination Code</i>
SMS-MO	<i>Mobile-Originated SMS: A message sent from the customer to a short number (e.g. 48XYZ) and forwarded by Pure Bros to the Provider</i>
SMS-MT	<i>Mobile-Terminated SMS: A message sent by the Provider via Pure Bros to the customer</i>
UDH	<i>User Data Header</i>
WAP	<i>Wireless Application Protocol</i>

4. Colour Conventions

To differentiate parameters within API calls, the following colour conventions apply:

-  **Red:** Mandatory parameters (case-sensitive)
 -  **Green:** Optional parameters (case-sensitive)
-

5. HTTP Requests

All API calls between Pure Bros and Providers use the GET method.

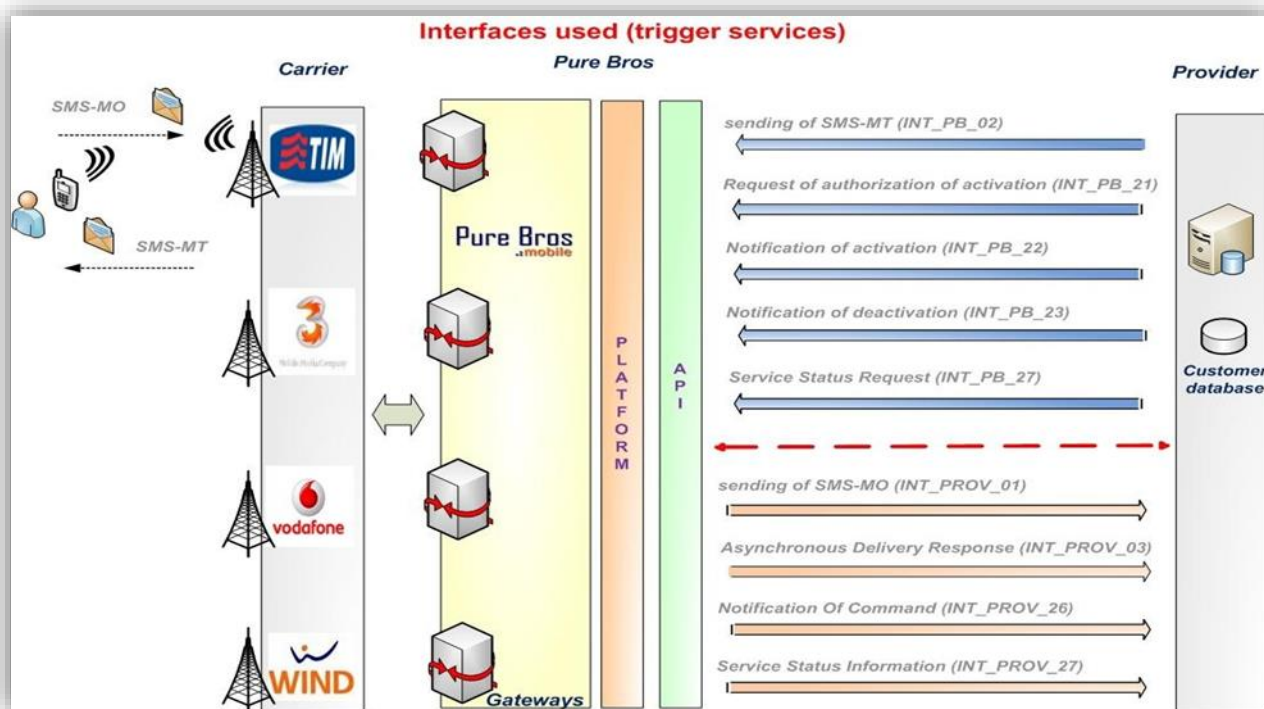
- Requests originated by Pure Bros → Sent as GET requests to the Provider's system.
 - Requests originated by the Provider → Sent as GET requests to the Pure Bros system.
-

6. Trigger Services

API Flow for Trigger Services

The diagram below illustrates the API interfaces used for a Trigger Service:

Fig.1- Interfaces used for a trigger service



 Pink arrows → Interfaces provided by the Provider, called by Pure Bros

 Blue arrows → Interfaces provided by Pure Bros, called by the Provider

Table 1 - Interfaces List.

Interface Name	API Code	Description	Owner
SMS-MT Delivery	INT_PB_02	Sends SMS-MT (free/charged)	Pure Bros
Activation Authorisation Request	INT_PB_21	Requests authorisation from the carrier to subscribe the customer	Pure Bros
Service Status Request	INT_PB_27	Queries Pure Bros for customer status	Pure Bros
SMS-MO	INT_PROV_01	Forwards SMS-MO from the customer to the Provider	Provider
Asynchronous Delivery Response	INT_PROV_03	Sends MT delivery status to the Provider	Provider

Interface Name	API Code	Description	Owner
Command Notification	INT_PROV_26	Sends activation/deactivation/suspension commands to the Provider	Provider

 **Note:** URLs for these interfaces must follow specific naming conventions.

 The URLs of the interfaces exposed by the Provider must follow the naming convention outlined following: <http://ProviderPath/PureBrosLabel?parameters>

7. API Interfaces

Each API interface is detailed below, including request parameters and example responses.

7.1 SMS-MO Interface (INT_PROV_01)

When a customer sends an SMS-MO to a Short Code (e.g. 48XYZ), Pure Bros forwards the message to the Provider for processing.

Request Parameters:

plaintext

http://ProviderURL/SMS-MO?pb_id=UniqueMessageID&pb_datetime=YYYY-MM-DDhh:mm:ss&mo_carrier=CarrierName&mo_source=393356334455&mo_target=48XYZ&mo_body=MessageText

Response Codes:

- ✓ 0 → Successfully received by the Provider
- ✗ 1 - <ErrorDesc> → Processing error

 **Example - Successful response:**

plaintext

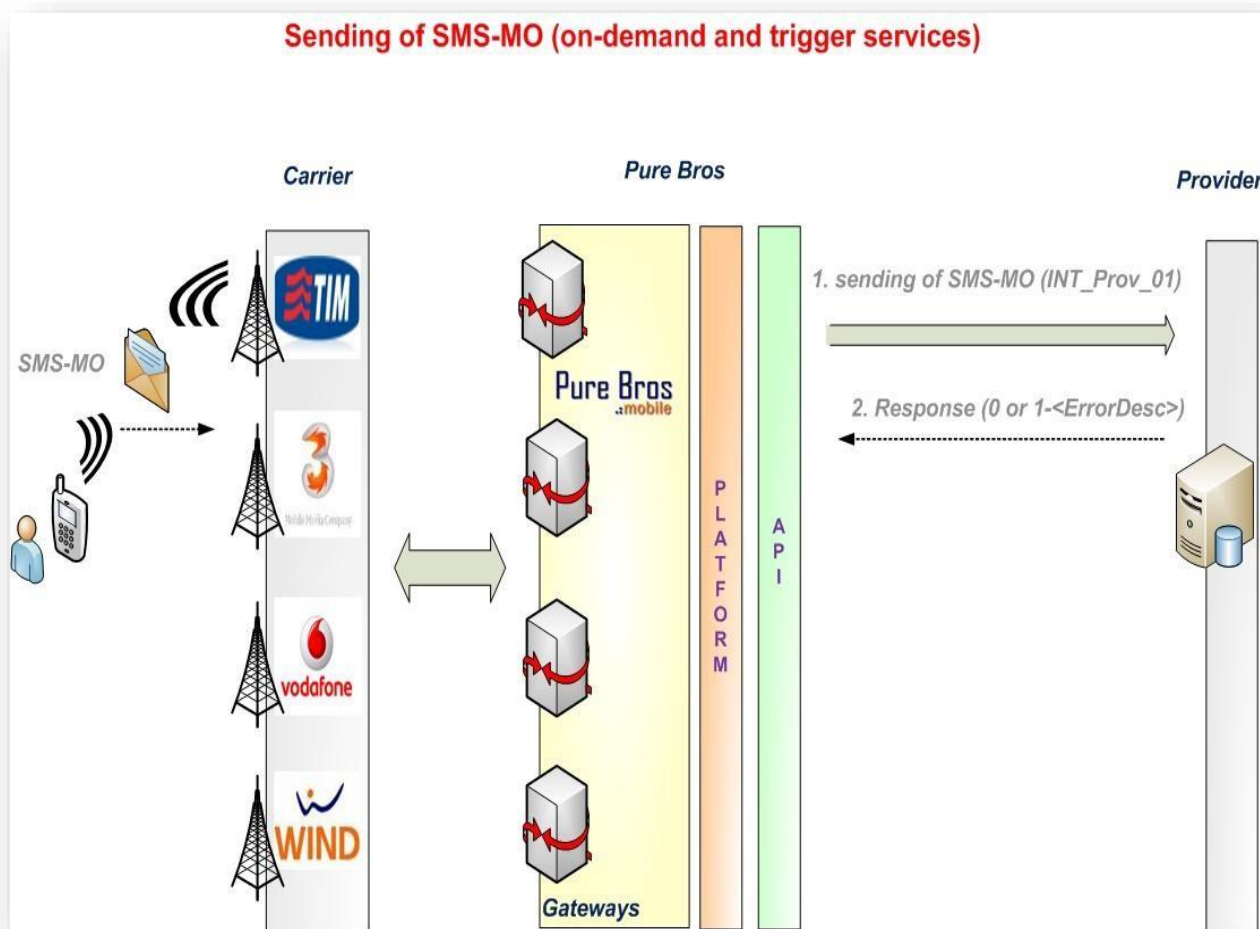
0

 **Example - Error response:**

plaintext

1 - DB Error Saving the received MO

Fig. 2 - SMS-MO Interface



Request Parameters

Name	HTTP Param	Description	Format	Occurence
PB.MessageID	pb_id	MO message unique identifier	VARCHAR (20)	<i>Mandatory</i>
PB.DateTime	pb_datetime	Timestamp of forwarding of the MO to Provider in Pure Bros system	CHAR (19) format YYYY-MM-DD hh:mm:ss	<i>Mandatory</i>
PB.Monitoring (Not used - ignore it)	pb_monitoring	Timestamp of forwarding of the MO to Provider in Pure Bros system	INTEGER 0 1	<i>Optional</i>

Name	HTTP Param	Description	Format	Occurrence
PB.Monitoring (Not used - ignore it)	pb_monitoring	If the value is "1", the message is a monitoring message sent from a Pure Bros monitoring device. In this case, the message body has a specific format and must be processed in a specific way. This parameter is deprecated	INTEGER 0 1	<i>Optional</i>
MO.Carrier	mo_carrier	Carrier that has received the SMS-MO sent from customer. This parameter can have one of these values: Tim/VodafoneMam/Vodafone/Wind/H3g.	VARCHAR (20)	<i>Mandatory</i>
MO.Source	mo_source	Customer's MSISDN. The format must be CC NDC SN where CC field (Country Code) for Italy is 39 (e.g.393356334455 is a valid MSISDN)	VARCHAR (40)	<i>Mandatory</i>
MO.Source.Scr amble (Not used - ignore it)	mo_source_scr amble	Carrier that has received the SMS-MO sent from customer.	INTEGER (0 1)	<i>Optional</i>
MO.Target	mo_target	ShortNumber 48xyz to which the MO was sent from customer	VARCHAR (15)	<i>Mandatory</i>
MO.MessageB ody	mo_body	which the MO was sent from customer	VARCHAR(1 60)	<i>Mandatory</i>
MO.DateTime	mo_datetime	The timestamp during which the MO was received from Carrier. This value will be the same of pb_datetime if the Carrier doesn't provide it	CHAR (19) Format YYYY- MM- DD hh:mm: ss	<i>Mandatory</i>
MO.Charged(Not used - ignore it)	mo_charged	If the MO was charged the parameter will contain the value of the charging. If the MO was not charged, the value can be	FLOAT NNN.DD	<i>Optional</i>

Name	HTTP Param	Description	Format	Occurence
		empty or "0.0". This parameter is deprecated		

Request Example:

plaintext

http://ProviderInterfaceAddress/ProviderName/PureBrosLabel?pb_id=MessageUniqueId&pb_datetime=YYYYMMDD:hh:mm:ss&mo_carrier=CarrierName&mo_source=393356334455&mo_target= 48xyz&mo_body= TextOfMessage&mo_datetime=YYYY-MM-DD hh:mm:ss

Name	Description	Occurence
Result code	The code indicates if the Provider processed or not the request sent from Pure Bros	Persistently present
ErrorDesc	The code indicates the Provider didn't received the MO. It should be a short description of the error	Not always present

Provider Response:

- ✓ 0 → MO successfully received by the Provider
- ✗ 1 → MO hasn't been received successfully by Provider

📌 Example - Successful response:

plaintext

0

📌 Example - Error response:

plaintext

1 - DB Error Saving the received MO

7.2 SMS-MT Delivery Interface (INT_PB_02)

This interface allows the Provider to send both free and charged SMS messages. It also supports WAP-Push and SMS messages containing downloadable content links.

Name	HTTP Param	Description	Format	Occurence
MT.Username	mt_user	Username will be submitted by Pure Bros during integration phase	VARCHAR (30)	<i>Mandatory</i>
MT.Password	mt_pass	Password will be submitted by Pure Bros during integration phase	VARCHAR (40)	<i>Mandatory</i>
MT.Format	mt_format	Format of content: 0 = Text (used also for SMS with link) 1 = Binary (used just for wap- push) N = Carrier Specific Format	INTEGER Default: 0	<i>Optional</i>
MT.UDH	mt_udh	If Format is binary, indicates whether the message has or not an UDH: 0 = without UDH 1 = with UDH	INTEGER Default: 0	<i>Optional</i>
PB.MOID	pb_moid	The identifier is associated with a previous MO sent from Pure Bros to Provider	VARCHAR (20) Default: empty	<i>Optional</i>
PB.CCID	pb_ccid	Identifier to the CreditCheck procedures. This parameter is used only in Wapbilling flow	VARCHAR (100)) Default: 0	<i>Optional</i>

MT.ContentID	mt_id	A unique identifier used for identify the specific transaction	VARCHAR (30)	<i>Mandatory</i>
MT.Source	mt_source	ShortNumber from which the SMS-MT must be sent. Usually is the same of Mo.Target value. Otherwise Pure Bros during integration phase will communicate the value that must be used	VARCHAR (15)	<i>Mandatory</i>
MT.Target	mt_target	Customer's MSISDN. The format must be CC NDC SN where CC field (Country Code) for Italy is 39 (e.g. 393356334455 is valid MSISDN)	VARCHAR (20)	<i>Mandatory</i>
MT.Carrier	mt_carrier	Carrier to which the SMS-MT must be sent. These exact values are released by Purebros and	VARCHAR (30)	<i>Mandatory</i>

Request Example:

plaintext

http://itn.purebros.com/cprovider/cdeliver?mt_user=ProviderUser&mt_pass=ProviderPassword&mt_format=0&mt_id=MessageUniqueId&mt_source=48XXX&mt_target=3934960445053&mt_bodycount=1&mt_body1=MessageText&mt_servtype=3333333333&mt_rmode=0&mt_reqsources=SMS&mt_carrier=CarrierName

Request Parameters:

Result Code	Description	Occurrence
-------------	-------------	------------

Result code	The code indicates if the MT has been accepted (Result code =0) or not (Result code <>0) from Pure Bros (see Appendix C in order to consult the list of Result code).	Persistently present
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✓ If Result Code = 0 (Asynchronous mode); Pure Bros has accepted the request and will send the MT to the carrier.

⚡ If Result Code = 0 (Synchronous mode); Pure Bros has accepted the request and will process it accordingly.

✗ If Result Code ≠ 0; The Provider did not include some parameters in the URL, or the IP, user, or password is incorrect. Typically, in this case, the Result Codes sent by Pure Bros are 98, 2,

Response Codes:

- ✓ 0 → Request accepted by Pure Bros
- ✗ 98 → Invalid parameters or incorrect authentication details

📌 Example - Success:

```
plaintext
```

```
0
```

📌 Example - Rejection:

```
plaintext
```

```
98
```

7.3 Asynchronous MT Delivery Response (INT_PROV_03)

This interface enables the Provider to receive the delivery status of a previously sent SMS-MT in asynchronous mode.

Request Parameters:

Name	HTTP Param	Description	Format	Occurrence
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MT.MessageID	mt_id	MT message unique identifier defined from Content Provider when sends the delivery request. This parameter is used for identify the specific transaction.	VARCHAR (30)	Mandatory
AR.ErrorCode	ar_errorcode	The code indicates the result of a previous request of MT delivery (see Appendix C in order to consult the list of Result code).	INTEGER	Mandatory

Request Example

```
plaintext
```

```
http://ProviderInterfaceAddress/ProviderName/PureBrosLabel?mt\_id=MessageUniqueID&ar\_errorcode=0
```

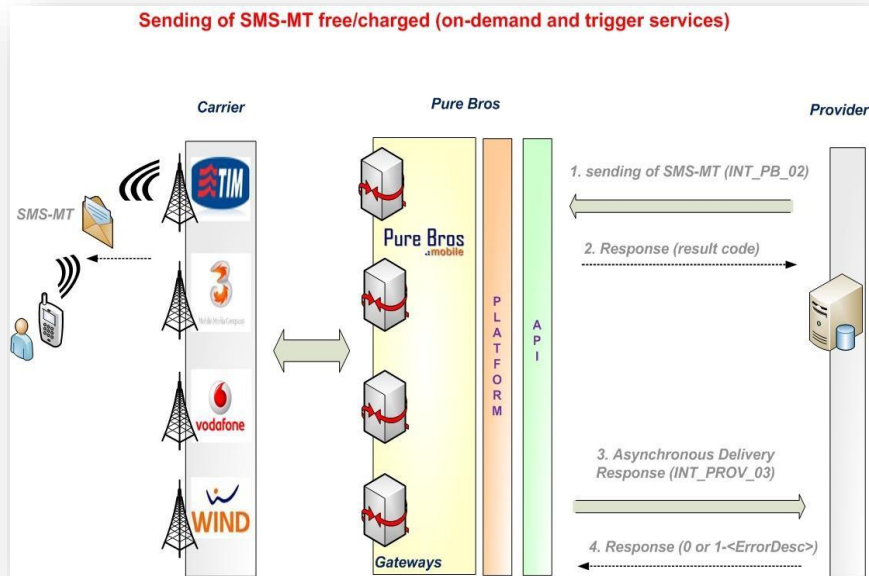
Possible Results:

- ✗ The SMS-MT has been rejected by the carrier (**ar_errorcode** ≠ 0)
- ✓ The SMS-MT has been successfully delivered by Pure Bros to the carrier (**ar_errorcode** = 0)

Response Parameters:

Name	Description	Occurrence
Result code	The code indicates if the Provider processed or not the request sent from Pure Bros	Persistently present
ErrorDesc	It describes why the notification of delivery wasn't processed from Provider	ErrorDesc

Fig. 3 - SMS-MTDelivery and Asynchronous Delivery Response interfaces



Request Parameters:

Parameter	HTTP Param	Description	Format	Occurrence
MT Message ID	mt_id	Unique identifier for the MT message, assigned by the Provider when sending the delivery request.	VARCHAR (30)	<i>Mandatory</i>
Error Code	ar_errorcode	Code indicating the outcome of the MT delivery request. (Refer to Appendix C for the list of response codes)	INTEGER	<i>Mandatory</i>

Request Example:

```
plaintext
```

```
http://ProviderURL/AsynchronousDeliveryResponse?mt_id=MessageUniqueId&ar_errorcode=0
```

Response Cases:

- SMS-MT successfully delivered to the carrier → **ar_errorcode** = 0
- SMS-MT rejected by the carrier → **ar_errorcode** ≠ 0

Response Format:

Response	Description
0	Notification successfully received by the Provider
1 - <ErrorDesc>	Notification processing failed (with an error description)

 Example - Success:

plaintext

0

 Example - Failure:

plaintext

1 - DB Error updating MT status

7.4 Request for Activation Authorization (INT_PB_21)

This interface is used by the Provider to request authorisation from the mobile carrier before subscribing a customer to a trigger (push) service. The subscription process varies depending on the customer's acquisition channel (e.g. SMS, WAP, WEB).

Regardless of the channel used, the Provider must invoke a request for authorization of activation. The subscription is permitted only if the synchronous response is positive, providing a specific authorization ID (authID). Otherwise, the Provider is not authorized to subscribe the customer.

There are cases where the Provider cannot subscribe a customer. In these cases, Pure Bros returns specific permanent error codes. The table below details these codes and the corresponding actions the Provider must take (refer to Appendix C for a complete list of Result Codes).

Result Code	Action
26, 36, 64	Retry of authorization request is forbidden. The customer cannot be subscribed.

 Depending on the channel used by the customer, the request must be made at a specific step in the flow. Typically:

- **SMS Channel:** The Provider must initiate the request only if the customer has sent an SMS-MO with the correct activation syntax. The syntax and alias must be

agreed upon with PB in a "Service Description" document submitted for carrier approval. If the syntax is incorrect, the Provider must send a support message using the MT Delivery interface.

- **WAP Channel:** Depending on the carrier service, if the customer subscribes via WAP, the Provider must integrate with the carrier's specific iframe platform (contact Pure Bros support for details).
- **WEB Channel:** A verification mechanism ensures that the person browsing is the SIM (MSISDN) owner:
 1. The customer enters their MSISDN in the designated web page and selects their carrier.
 2. The Provider sends a PIN code.
 3. The customer returns to the web page and selects a content item or confirms the subscription.
 4. Only after this process is completed can the Provider submit the request.

 Regardless of the channel used, the authorization request is always the first step when a customer subscribes to a service.

Request Parameters

Name	HTTP Param	Description	Format	Occurrence
MT.Username	mt_user	Username provided by Pure Bros during integration	VARCHAR (30)	Mandatory
MT.Password	mt_pass	Password provided by Pure Bros during integration	VARCHAR (40)	Mandatory
MT.Target	mt_target	Customer's MSISDN. The format must be CC NDC SN where CC field (Country Code) for Italy is 39 (e.g. 393356334455 is a valid MSISDN)	VARCHAR (20)	Mandatory
MT.Carrier	mt_carrier	Carrier to which the request of authorization	VARCHAR (30)	Mandatory

		must be sent. These exact values are released for CSP involved on integration phase		
MT.ServiceType	mt_servtype	Specific service type for the request, provided by Pure Bros before service launch	INTEGER	Mandatory
MT.ParamCount	mt_params	Number of parameters to be sent. If set to 0, the following parameters can be omitted. Default: 0	INTEGER	Mandatory
MT.ParamName	mt_paramNameN	Name of parameter N (sequential numbering required)	VARCHAR(255)	Mandatory
MT.ParamValue	mt_paramValueN	Value corresponding to parameter N	VARCHAR(255)	Mandatory

 **Mandatory Parameter:** The "channel_in" parameter must be included in MT.ParamName. This parameter specifies the provisioning channel used by the customer to request the service (refer to Appendix F for possible values).

Request Example:

plaintext

http://itn.purebros.com/cprovider/ActivationAuthorisation?mt_user=ProviderUser&mt_pass=ProviderPassword&mt_target=393356331133&mt_carrier=CarrierName&mt_servtype=11111111&mt_params=1&mt_paramName1=channel_in&mt_paramValue1=SMS

Response	Description	Occurence
Result code	The code indicates the result of a previous request of authorization (see Appendix C to consult the list of Result code).	Persistently present
Authorization_id	Authorization ID. Is an unique identifier for each subscription. Pure Bros gives this value in sync	Present just if the Result code = 0 (the

Response	Description	Occurence
	mode during authorization phase (in the response related at a request of authorization).	authorization request processed with success)

Response Codes:

- ✓ 0 → Authorisation granted, an Authorisation ID is provided
- ✗ 60 → Customer already active
- ✗ 26, 36, 64 → Subscription denied permanently

📌 Example - Successful response:

plaintext

0

12345

📌 Example - Rejection:

plaintext

60

7.5 Command Notification (INT_PROV_26)

This interface is used by Pure Bros to send deactivation commands received from the carrier to the Provider. The Provider must immediately process the deactivation and must not call the Deactivation Notification API (INT_PB_23) since the customer has already been unsubscribed.

Request Parameters:

Name	HTTP Param	Description	Format	Occurence
MT.Username	mt_user	Username will be submitted by Pure Bros during integration phase	VARCHAR (30)	Mandatory
MT.Password	mt_pass	Password will be submitted by Pure Bros	VARCHAR (40)	Mandatory

		during integration phase		
MT.Carrier	mt_carrier	Carrier to which the customer appertains. These exact values are released by purebros to CSP involved during integration phase	VARCHAR (30)	Mandatory
MT.Target	mt_target	Customer's MSISDN. The format must be CC NDC SN where CC field (Country Code) for Italy is 39 (e.g. 393356334455 is a valid MSISDN)	VARCHAR (20)	Mandatory
MT.ServiceType	mt_servtype	Pure Bros will communicate the service type that the Provider can use in the http request before the launch of the service.	INTEGER	Mandatory

Request Example

plaintext

```
http://ProviderURL/CommandNotification?mt_carrier=CarrierName&mt_op=1&mt_target=3356655333&mt_servtype=5555555555&mt_authID=12131415&mt_paramName1=sms_line&mt_paramValue1=48XXX&mt_paramName2=reason&mt_paramValue2=USER_REQUEST&mt_paramName3=channel_in&mt_paramValue3=WEB
```

Response Codes:

- ✓ 0 → Command successfully processed
- ✗ 1 - <ErrorDesc> → Error processing the command



Example - Successful response:

plaintext

0



Example - Error response:

plaintext

1 - DB Error updating MT status

7.6 Service Status Request (INT_PB_27)

This interface allows the Provider to query Pure Bros for the customer's subscription status and retrieve the Authorisation ID issued during activation.

Request Parameters:

plaintext

http://itn.purebros.com/cprovider/ServiceStatus?mt_user=ProviderUser&mt_pass=ProviderPassword&mt_carrier=CarrierName&mt_target=393334552021&mt_servtype=6666666666

Response Codes:

-  0 → Query successful, returns customer status and (if active) the Authorisation ID
-  98 → Invalid parameters

 Example - Customer active:

plaintext

0

0

123456 (authorization code)

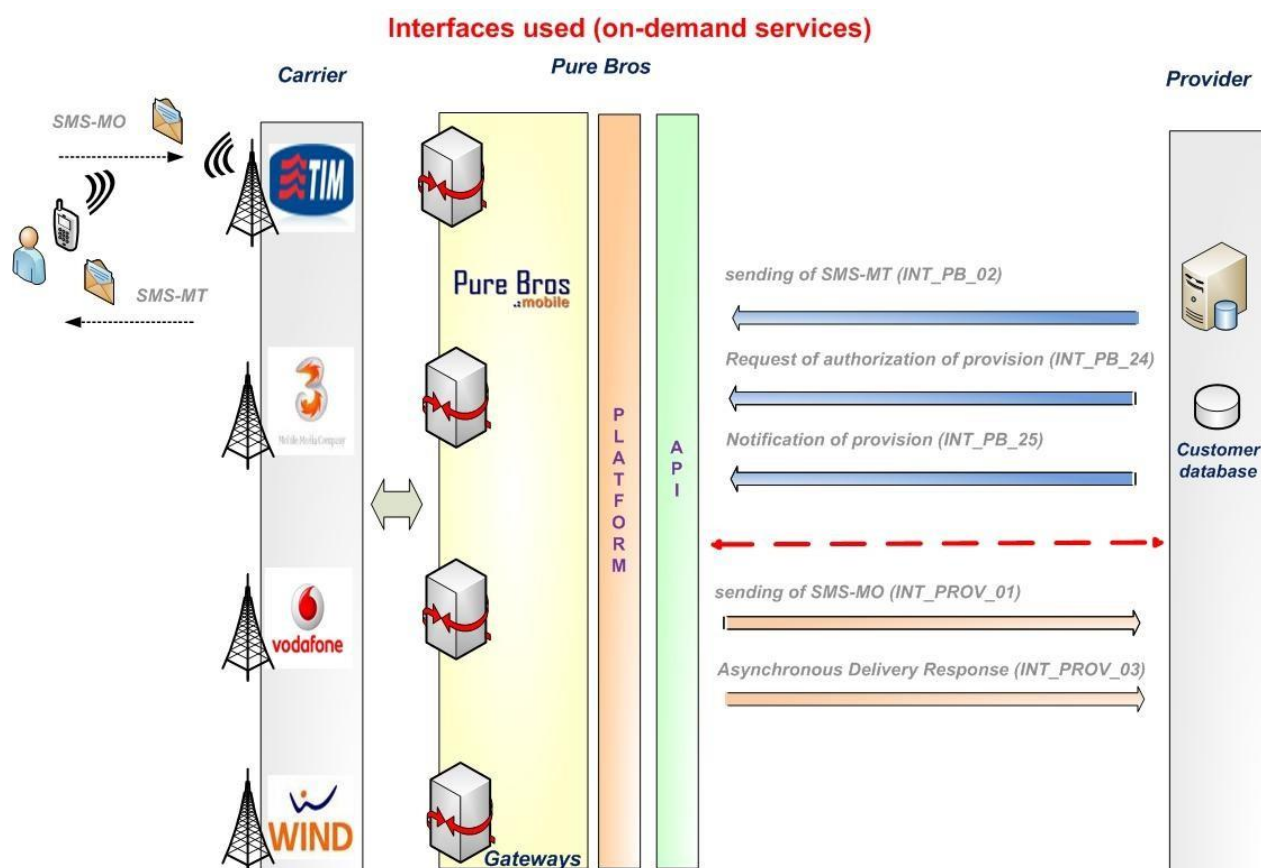
 Example - Customer deactivated:

plaintext

0

1

8. On-Demand Services



The diagram illustrates all interfaces used for an on-demand service.

- Pink arrows represent interfaces exposed by the Provider and invoked by Pure Bros
- Blue arrows indicate interfaces exposed by Pure Bros and invoked by the Provider

This chapter covers the newly introduced Request for Authorization of Provision (INT_PB_24), which is mandatory only for TIM. Other interfaces relevant to on-demand services have already been discussed in the previous chapter:

- INT_PROV_01
- INT_PROV_03
- INT_PB_02

Provisioning Flow for On-Demand Services

For on-demand services, the Provider must follow a specific provisioning flow:

1. Request Authorisation for Provisioning (AuthID) → INT_PB_24
2. Send Content/Billing to the Customer → INT_PB_02

 **Note:** INT_PB_24 is mandatory only for TIM.

Appendices

A: Pure Bros Network Parameters

Pure Bros will provide a separate document detailing its network parameters and those required from the Provider.

B: Binary Body Format

Binary content must be converted into a Hex Byte representation and concatenated into a single Hex String.

Example Conversion Table:

Byte	Hex Byte
10	0A
11	0B
112	70
254	FE
255	FF

C: Response Codes

A full list of response codes is available in Appendix C.

Code	Description
0	Operation executed successfully
1	Invalid Username/Password
2	Connection not allowed from source address
3	Permission denied using this operation
12	MT.Body# is out of MT.BodyCount range

Code	Description
13	ServiceType not set to the requested operation. Contact Pure Bros Contact
14	Invalid content source
21	Target phone number not covered by Pure Bros
22	Unauthorized Carrier
24	Target phone is blocked by blacklist
25	Retry not allowed
26	Invalid target on carrier
32	Problems connecting to carrier
33	Billing confirmation not received
34	Content not delivered: MT.ContentID already exists
35	KO - Retry allowed
36	Operation denied for carrier
37	Invalid operation
44	Msisdn deleted
51	The maximum service amount was already reached
53	The maximum charge retries was already reached
60	The TARGET is already active
63	Service temporary unavailable (SNA)
64	Service not found
65	The TARGET is already deactivated
91	System closed
96	Internal database access error

Code	Description
97	Timeout error
98	Error processing the request: invalid parameter(s)
99	Internal error processing the request

D: HTTP Protocol

All API requests adhere to the HTTP 1.1 protocol. Further details can be found in RFC 2616.